

Filippo Ficarra

Experience

- Jan 2026 – **Capacity, Starmind Team, Machine Learning Engineer, Zürich**
 Now **Expertise Graph & Recommendations:** Architecting an E2E pipeline using Docling to transform 1M+ unstructured PDFs into a concept ontology; implemented NER and vector clustering for autonomous node instantiation to power an expertise-based query recommendation model.
- Jan 2025 – **IBM Research, Research Scientist Intern, Zürich**
 July 2025 ○ **Multimodal Material Discovery:** Developed a Transformer-based architecture to identify molecular structures within complex IR mixtures. Engineered a high-throughput data pipeline that simulate 200B+ samples across diverse mixture settings and concentrations. Achieved 34% Top-1 accuracy, establishing the first task benchmark and outperforming human expert performance. Accepted as a workshop paper at **NeurIPS 2025**. 🌐 [1].
 ○ **Automated Catalyst Extraction:** Designed an automated pipeline using *Docling* and *LangChain* to extract structured Single Atom Catalyst data from PDF literature; fine-tuned IBM Granite 3.2 8B via LoRA to predict laboratory-ready synthesis procedures.
 ○ **Large-Scale Model Training:** Orchestrated supervised fine-tuning for Llama 3 models up to 405B parameters on 2 to 2,048 Nvidia GH200 GPUs at CSCS Alps; utilized *DeepSpeed ZeRO-3* for training efficiency and profiled scaling laws to secure a major research grant.
- Nov 2023 – **ETH Zürich, SwissAI Project (Rycolab & VILAB)**
 June 2024 Contributed to the design of a large-scale multimodal foundation model; built scalable data curation and distributed data-loading pipelines for training on the CSCS Alps cluster.
- Sep 2023 – **ETH Zürich, Teaching Assistant**
 Sep 2024 Assisted LLM and Information Security labs for 200+ students. Covered advanced prompting strategies and language model calibration, along with hands-on exploration of security vulnerabilities and attack techniques, including cryptographic attacks, LLM jailbreaks, adversarial attacks, formal verification with Tamarin Prover, and binary exploitation.
- Mar 2022 – **ORBYTA TECH, Backend Engineer Intern, Torino**
 Jun 2022 Developed a scalable REST API for a B2B clothing platform. Made use of *C#* and *MySQL*.
- Nov 2021 – **JEToP, IT Consultant, Torino**
 Aug 2022 Built a cross-device podcast application using *React Native* and *TypeScript*.

Education

- Sep 2022 – **ETH Zürich, MSc in Computer Science**
 Nov 2024 Major in **Machine Intelligence**, Minor in **Information Security**.
 ○ Master Thesis (Rycolab): Investigated LLM word acquisition vs. humans using surprisal-based metrics. Accepted at **NAACL 2025** (Long Paper). 🌐 [2].
- Sep 2019 – **Politecnico di Torino, BEng in Computer Engineering, Final Grade: 110/110 cum laude**
 Jul 2022 Full Merit Scholar: Selected for the **top 1%** excellence track "Percorso Giovani Talenti".

Publications

- [1] Marvin Alberts[†], **Filippo Ficarra[†]**, and Teodoro Laino. [Language Model Enabled Structure Prediction from Infrared Spectra of Mixtures](#). In *AI for Accelerated Materials Design - NeurIPS 2025*, 2025.
- [2] **Ficarra, Filippo**, Ryan Cotterell, and Alex Warstadt. [A Distributional Perspective on Word Learning in Neural Language Models](#). In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL)*, pages 11184–11207, Albuquerque, New Mexico, April 2025.

Skills

ML & AI	LLMs, Multimodal Models, Distributed Training, Vector Search	Languages	Python, JavaScript, Java, C, C#, SQL, Bash
Frameworks	PyTorch, DeepSpeed, LangChain, Hugging Face	DevOps	Docker, Kubernetes, AWS, GCP, Slurm, Jenkins